Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/23/2008
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Project Title:	Direct Visualization and Light Scattering Studies of the Structure, Dynamics and Rheology of Anisometric Colloidal Suspensions and Gels		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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Flight Program:			
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Key Personnel Changes/Previous PI:	0		
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Task Description:	This project seeks to quantify the structure and dynamics of colloidal suspensions comprised of anisometric (non-spherical) colloids by investigating a unique set of materials developed in the course of NASA-sponsored resarch. Methods include confocal microscopy and light scattering. There is an unfulfilled need to better understand the fundamental fluid physics that leads to anisotropic excluded volume effects on the structure, dynamics and rheology of anisometric colloidal suspensions. This project addresses this unfulfilled need. Complex fluids consisting of submicron colloids dispersed in simple fluids are key materials in the industrial sectors of ceramics, composites, automotive, pharmaceutical, coatings and personal care. Although the chief advances in fundamental research have been for materials comprised of spherical colloids, applications involving anisometric colloids are important and growing. For example, anisometric colloids with rod or disk-like shapes are key components of materials such as magnetic media for computer		

	data storage, polymer/clay nanocomposites for automotive and aerospace applications. Potential applications of ordered rod suspensions in sensors and optical materials can also be envisaged. Understanding of anisometric colloid structure is useful for the development of new materials with anisotropic optical, permeability and mechanical properties for applications such as filtration, purification and sensing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Technological applications in the development of materials with anisotropy in optical, permeability and mechanical properties benefits the nation.
Task Progress:	Science Progress and Status Confocal microscopy of colloidal rod suspensions: A) Effect of temperature on sedimentation of PMMA-g-PDMS colloidal rods in hexane is being studied to search for unusual liquid crystal structures. Colloidal rods are synthesized by uniaxial extension of microspheres as described in our Langmuir publication. Sedimentation of colloidal particles is primarily controlled by Peclet number, which is ratio of settling forces to diffusive forces. We have observed different kinds of orientational ordering as a function of temperature. The study of sediments in which the director twists with distance from the microscope are of particular interest. Structures at different temperatures (24°C and 10°C) were identified as nematic and cholesteric liquid crystalline phases, respectively. It was observed that initial particle concentration plays a significant role in defining sediment thickness for nematic phases and pitch for cholesteric phases. Also, it was found that there is a minimum initial particle concentration below which no ordering is observed. This minimum concentration is also a function of temperature. We are in process of developing a phase diagram to better understand the transition between these different liquid crystal orderings. B) We are conducting sedimentation experiments at high solvent viscosities (˜ 100 cp) to understand arrangement of rods during sedimentation. High solvent viscosity retards both translation and rotational motion of particles. The retardation provides sufficient time for image acquisitoin using confocal laser scanning microscopy (CLSM). Initial experiments have shown that with sufficient viscosification we are able to observe the rotation and translation motion of individual rods under dilute conditions. We are in the process of extending these methods to interrogate dynamics of concentrated rod suspensions. C) We have synthesized new PMMA microspheres, with grafting layer of UV curable PDMS. These particles when exposed to UV light bind with each other. The grafting layer is composed of two different PDMS molecules, methacryloxypropyl terminated PDMS (MW: 25000) and 3-acryloxy-2-hydroxypropyl PDMS (MW: 1000). 2-2-Dimethyl-2-phenylacetophenone (DMPA) is used as the initiating agent for UV reaction. These particles will be used to synthesize anisometric structures. D) We are developing a new method to fabrication anisotropic rods by manipulation of spheres in microfluidic channels to yield linear chains of particles that are then bonded to yield rigid rods. Reaction and control schemes are being developed to synthesize such anisometric colloidal chains by reaction of spherical particles. We have executed reactions between carboxyl and amine modified polystyrene spheres (CM-PS and AM-PS) in pursuit of this aim. In these syntheses, CM-PSs can be reacted with a molecule with two amine groups (Alexa fluor 488) in presence of initiator EDAC (1-ethyl-3-(3-dimethylaminopropyl)-carbodiimide in MES Buffer of pH 6.1. CM-PS and AM-PS colloids can be reacted in the presence of initiator EDAC at a pH of 6.1. Reaction of two AM-PSs is accomplished by means of a linker with two aldehyde groups (Gluteric Dialdehyde) in phosphate buffer (pH = 6.5). These reactions have all been carried out in bulk. We are in the process of executing these methods in narrow channels to yield linear chains of colloids. Light scattering and rheological studies of colloidal rod suspensions. A paper on the dynamic light scattering behavior of model, monodisperse boehmite colloidal rods of aspect ratio between 4-30 is in the final stages of preparation. A paper demonstrating the effect of bond angle rigidity on the non-linear rheology of dilute fractal gels of rods and spheres was accepted for publication in Journal of Rheology. We are preparing a manuscript that applies soft glassy rheology models to colloidal gels to understand their non-linear rheology. Plans for 2006 Our work will continue to focus on three areas: (1) Order/disorder transitions in rod suspensions. We will complete our measurement of the phase diagram quantifying the bounds of the region of the order/disorder orientation transition (nematic-like) that we discovered in fluorescent suspensions of colloidal rods. Our initial observation of cholesteric phases needs to be understood in terms of the underlying shape of the rods. We aim to develop novel methods to control the shape of the particles that are produced from our uniaxial deformation scheme. If novel phases are achieved, we will evaluate these materials for their technological potential as sensors and optical materials. (2) Novel anisotropic rods. We will collaborate with Professors M. Burns and J. Mirecki-Millunchick to execute covalent bonding reactions between spheres in microfabricated channels to yield rigid rods comprised of spheres with variable interactions. (3) Rheology of anisometric gels. We will continue to apply soft glassy rheology models to understand the unusual transient rheology of gels comprised of non-spherical particles. SUPPLEMENTAL INFORMATION FOR FINAL REPORTING: Summary: This project sought to quantify the structure and dynamics of colloidal suspensions comprised of anisometric (non-spherical) colloids by investigating a unique set of materials developed in the course of NASA-sponsored research. Complex fluids consisting of submicron colloids dispersed in simple fluids are key materials in the industrial sectors of ceramics, composites, automotive, pharmaceutical, coatings and personal care. Although the chief advances in fundamental research have been for materials comprised of spherical colloids, applications involving anisometric colloids are important and growing. For example, anisometric colloids with rod or disk-like shapes are key components of materials such as magnetic media for computer data storage, polymer/clay nanocomposites for automotive and aerospace applications. Potential applications of ordered rod suspensions in sensors and optical materials can also be envisaged. These materials derive their functionality from colloidal scale shape anisotropy. Thus, there is an unfulfilled need to better understand the fundamental fluid physics that leads to anisotropic excluded volume effects on the structure, dynamics and rheology of anisometric colloidal suspensions.

	In this project, we explored the regimes under which rods suspensions adopt liquid crystal-like ordering in sedimentation. We developed methods to interrogate the rod dynamics that lead to such ordering. We developed methods to quantify the translational and orientational diffusion of anisometric colloids in three dimensions. Our NASA supported work in rod assembly was profiled on the cover of Langmuir and AIChE J.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
Articles in Peer-reviewed Journals	Mohraz A, Solomon MJ. "Orientation and rupture of fractal colloidal gels during start-up of steady shear flow." Journal of Rheology 2005 May-Jun;49(3):657-81. http://dx.doi.org/10.1122/1.1895799 , May-2005
Articles in Peer-reviewed Journals	Mohraz A, Solomon MJ. "Direct visualization of colloidal rod assembly by confocal microscopy." Langmuir. 2005 Jun 7;21(12):5298-306. PMID: 15924453 , Jun-2005
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Journal/Magazine covers	Mohraz A, Solomon MJ. "Cover image in journal Langmuir for article, “Direct visualization of colloidal rod assembly by confocal microscopy." Langmuir. 2005 Jun 7;21(12):5298-306. PMID: 15924453 , Jun-2005
Presentation	Mohraz, A. and M.J. Solomon "Structural evolution and non-linear rheology of weak colloidal gels during start-up of steady-shear flow" Annual Meeting of the Society of Rheology, Lubbock TX Feb-2005
Presentation	Mohraz, A. and M.J. Solomon "Orientational order in colloidal rod suspensions by sedimentation" ACS National Meeting, San Diego CA Mar-2005